

Water Through Rock Lab

I can follow or direct a fixed water-passage method, record what reaches the collector and make a careful comparison of supplied samples.

SEAL

water must not bypass the sample

FOLLOW

same stated lab method

RECORD

collector result in mL

[Open the original interactive lesson](#)

What success looks like

- Follow the fixed volume, time and sealed-route instructions safely.
- Record surface observations and water collected below in mL.
- Compare the supplied samples under this method without claiming waterproof forever.

Retrieve and reconnect

- Last lesson: what observation directly showed passage through?
- Last lesson: why did an edge leak fail the method check?
- Last week: state one sample-specific conclusion rule.
- Lead-in: why must the collar stay sealed?

Water appears in the tray, but the collar leaked around the sample edge. What should the lab team record?

Think first. Point, say, sign or write your choice. Give one reason.

A. Count all tray water as passed through the rock.

B. Mark the run invalid because water bypassed the sample; stop and use the repaired or fixed-data route.

C. Guess how much probably went through and keep the result.

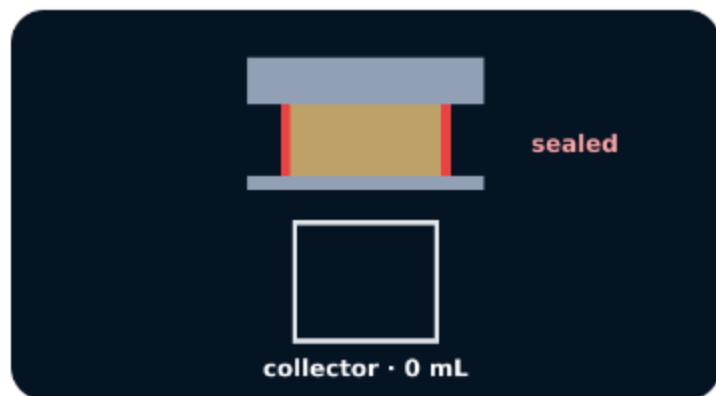
Look closely · what is the evidence?

Observation 1: water is visible outside the sealed test area.

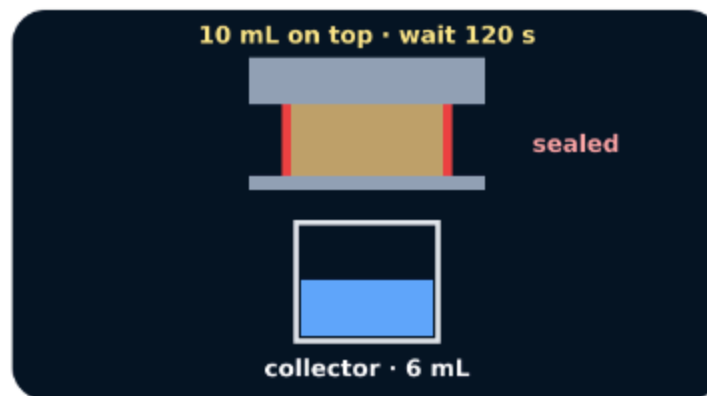
Observation 2: the collector and spill tray are not the same place.

Decision: valid passage evidence must reach the collector through the prepared sample.

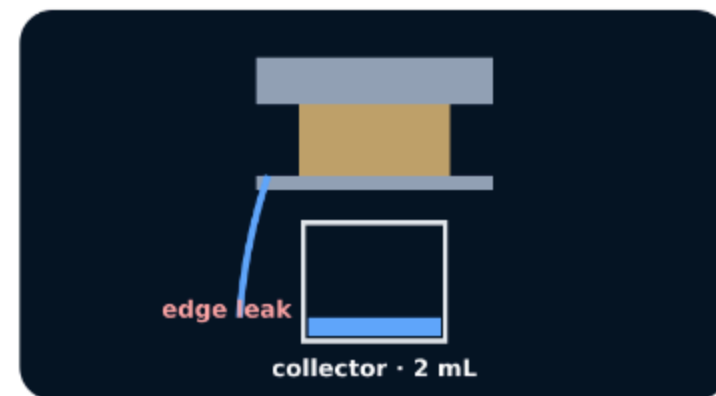
Look closely · what is the evidence?



A sealed sample above a collector



Water collecting below the sample



An edge leak spoiling a result

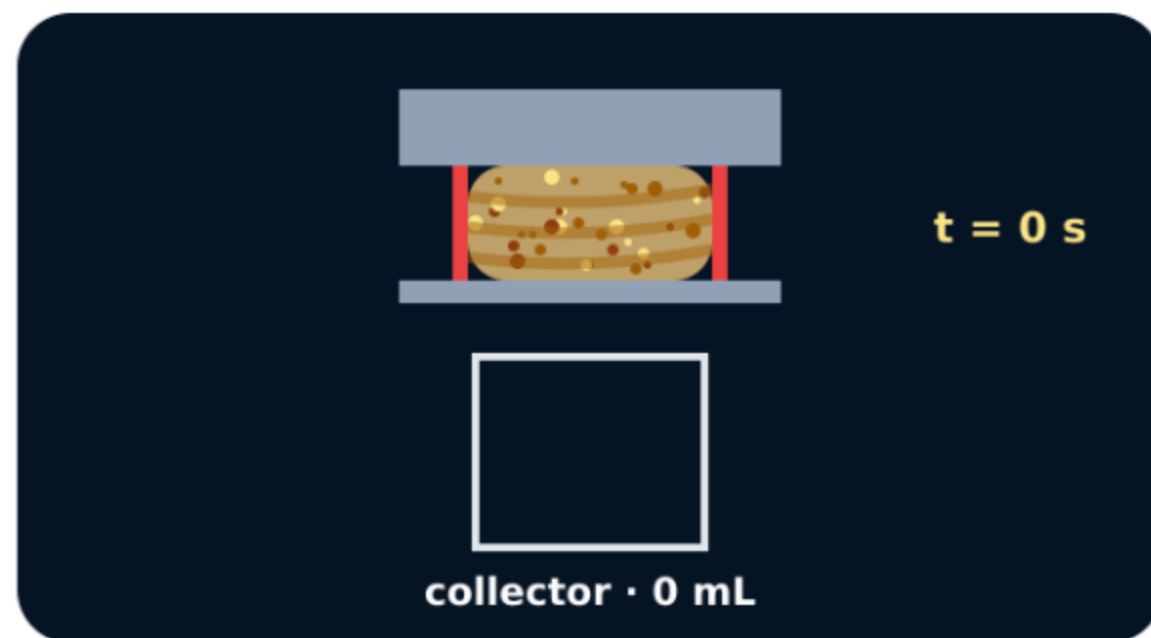
Words we will use

- permeability — how water passes through
- volume — amount of liquid
- valid — the method worked as stated

Watch the model

- Staff check the stable tray, pre-inspected matched pieces, sealed collar and clear collector. Pupils may operate only agreed safe steps or direct an adult.
- Add the fixed 5 mL with a needle-free tool, start the 120-second timer and observe top, underside and collector without moving the rig.
- Record the amount collected below in mL and note surface-only, absorbed or passed-through evidence. Record a fault instead of inventing a result.

Watch the model



Sealed water bench with collector

Sample A collects 0.5 mL and Sample B collects 3.0 mL after the same 5 mL and 120-second method. Which comparison is supported?

- A. More water passed through supplied Sample B under this method.
- B. Sample A is waterproof forever.
- C. Every rock like B always passes exactly 3.0 mL.

Commit to an answer. Show the evidence you used. Listen, then revise if needed.

Build the explanation

- Compare collected-water values only after checking that both runs were valid. A lower value means less passage was observed in the stated time, not proven permanent waterproofing.
- Notice what the fixed method tried to hold steady: starting volume, time, collar area and prepared thickness. W13 will name and test these method controls directly.
- End with one careful sentence: Under this method, more water passed through supplied Sample _____. One result set is useful evidence, but samples and repeats can vary.

Find and repair the misconception

Lab glitch: any water in the tray counts as water through the rock.

A. Keep it: spills, leaks and collector water all mean the same thing.

B. Repair it: count only water reaching the collector through the sealed sample route; mark leaks as faults.

C. Repair it: estimate the leak and add a guessed collector result.

Water-passage lab console

Route each lab record to VALID A, VALID B, METHOD FAULT or CAREFUL COMPARISON.

- Check seal, volume and time on the record.
- Place valid results with the sample code.
- Move faults away from the comparison.
- Freeze the evidence and state a bounded comparison.

Use the numbered cards in your pupil pack. Take turns to choose, justify and check. You may direct a partner to move a card.

Choose a group · defend your placement

- VALID · SAMPLE A
- VALID · SAMPLE B
- METHOD FAULT · DO NOT COMPARE
- CAREFUL COMPARISON

Use the frozen records to say what passed through, which run was valid and what the result does not prove.

Your independent mission

Follow or direct the fixed water-passage lab for supplied Samples A and B, or analyse the fixed equivalent records, then give a bounded comparison.

Record: Method checklist, surface/underside observation, collector values with mL, valid/fault decision, bounded comparison and one method-control question for W13.

Choose your support and challenge

- Supported: Sequence the fixed method, place A and B result cards and complete an oral, pointed or scribed comparison.
- Standard: Record both valid results, reject one fault record and explain the sample comparison with a time limit.
- Stretch: Evaluate how thickness, seal or natural variation could affect the comparison and propose one control to investigate in W13.

Show what you know

Give the comparison and explain why it does not mean waterproof forever.

Point to, read or explain the evidence you made. Choose one next step after the adult has listened.